Modelling of Dynamical Systems

3D printer

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Introduction

The mini-project involves the modelling of a 3D-printer in three domains: physical, discrete, and hybrid.

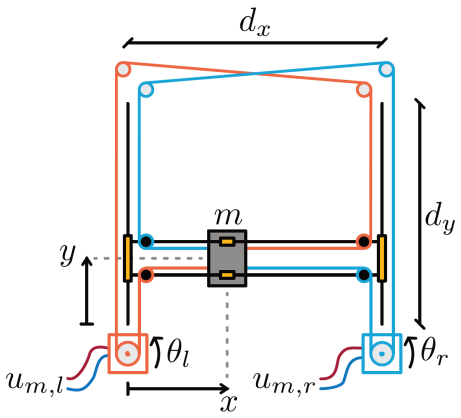


Figure 1: The schematic view of the 3D-printer.

In Figure 1 the schematic overview with the associated variable names and orientations is shown. The orientation and names of the corresponding physical quantity are respected in the derivation of the models. The mandatory variable names are given in Table 1. The formulas contain auxiliary variables which may not be given in this table.

Table 1: Variables and parameters of the 3D-printer.

Name	**Symbol**	**Value**	**Unit**
Motors and print head			
Motor angle left	θ_l		rad
Motor voltage left	$u_{m,l}$		V
Motor current left	$i_{m,l}$		A
Motor radius	r_m	11	mm
Motor inductance	L_m	10	mH
Motor resistance	R_m	90	Ω
Motor inertia	J_m	$8 \cdot 10^{-6}$	$\mathrm{kg\,m^2}$
Motor constant	K_m	0.07	N m/A
Tool head position	x, y		m
Tool head mass	m	60	g
Nozzle and thermal parameters			
Nozzle voltage	u_n		V
Nozzle current	i_n		A
Nozzle capacitance	C_n		F
Nozzle thermal capacity	$C_{t,n}$		J/K
Nozzle thermal resistance	$R_{t,n}$	9	K/W
Nozzle resistance	R_n	8	Ω
Nozzle temperature	T_n	240	°C
Environmental temperature	T_e	300	K
Bearing friction			
Rolling bearing friction	μ_r	0.003	N s/m
Linear bearing friction, x-direction	$\mu_{l,x}$	0.002	N s/m
Linear bearing friction, y-direction	$\mu_{l,y}$	0.001	N s/m
Print bed and geometry			
Print bed temperature	T_b	105	°C
Print area dimension, x-direction	d_x	29	cm
Print area dimension, y-direction	d_y	29	cm

Right motor variables are analogous to those of the left motor. Empty value cells indicate variables or unspecified parameters. The unit of motor inertia is not specified in the assignment.

1 Physical Modelling

Question 1 Kinematics

The displacement of the belt by the driving motors having a radius of r_m can be expressed by the following equation

$$\Delta s = r_m \Delta\theta. \tag{1}$$

This leads to the following equations for the left and right motor being described by

$$r_m \Delta\theta_l = \Delta x + \Delta y, \qquad r_m \Delta\theta_r = \Delta x - \Delta y. \tag{2}$$

Adding and subtracting the two belt equations gives the displacements Δx and Δy, respectively:

$$\begin{aligned} 2\Delta x &= r_m \Delta\theta_l + r_m \Delta\theta_r & \quad 2\Delta y &= r_m \Delta\theta_l - r_m \Delta\theta_r \\ &= r_m(\Delta\theta_l + \Delta\theta_r), & &= r_m(\Delta\theta_l - \Delta\theta_r), \\ \Delta x &= \frac{r_m}{2}(\Delta\theta_l + \Delta\theta_r). & \Delta y &= \frac{r_m}{2}(\Delta\theta_l - \Delta\theta_r). \end{aligned} \tag{3}$$

When the reference is taken to be zero (i.e., $x_0, y_0, \theta_{l,0}, \theta_{r,0} = 0$), the prefix Δ can be discarded. The earlier derived equations simplify to

$$x = \frac{r_m}{2}\left(\theta_l + \theta_r\right), \qquad y = \frac{r_m}{2}\left(\theta_l - \theta_r\right). \tag{4}$$

The velocities can be found by differentiation

$$\dot{x} = \frac{r_m}{2}\left(\dot{\theta}_l + \dot{\theta}_r\right), \qquad \dot{y} = \frac{r_m}{2}\left(\dot{\theta}_l - \dot{\theta}_r\right). \tag{5}$$

The rotational speed is given by the derivative of the angle (i.e., $\dot{\theta} = \omega$). Therefore, the velocities can alternatively be denoted by

$$\dot{x} = \frac{r_m}{2}\left(\omega_l + \omega_r\right), \qquad \dot{y} = \frac{r_m}{2}\left(\omega_l - \omega_r\right). \tag{6}$$

For completeness, the accelerations are the second derivatives

$$\ddot{x} = \frac{r_m}{2}\left(\dot{\omega}_l + \dot{\omega}_r\right), \qquad \ddot{y} = \frac{r_m}{2}\left(\dot{\omega}_l - \dot{\omega}_r\right). \tag{7}$$

Question 2 Dynamics

The model of the DC motor coupled to a disk with an inertia is shown in Figure 2.

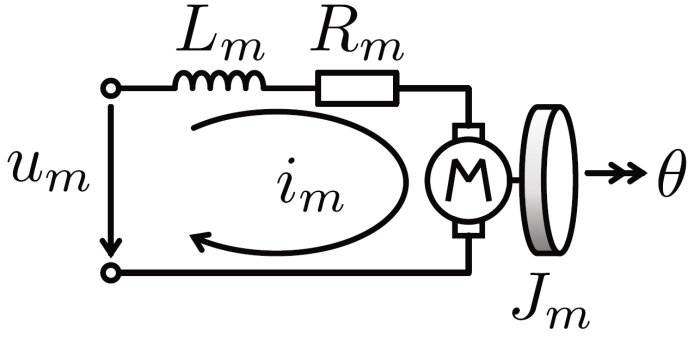


Figure 2: The proposed DC motor modelled as an LR circuit.

2.1 Electrical dynamics

The torque of the motor is related to the current flowing through the motor $\tau_m = K_m i$. Applying Kirchhoff's voltage law to the schematic of Figure 2 leads to*

$$u_m = u_{L_m} + u_{R_m} + u_{\text{EMF}}, \tag{8}$$

$$u_m = L_m \frac{di_m}{dt} + R_m i_m + K_m \dot{\theta}, \tag{9}$$

$$L_m \frac{di_m}{dt} = -R_m i_m - K_m \omega + u_m. \tag{10}$$

2.2 Mechanical dynamics

The printer head is modelled as a mass. The belts driven by the motors exert forces on the mass denoted by F_x and F_y in Figure 3. Friction due to the linear bearings causes an opposite force acting on the mass. This effect is modelled by the $\mu_{l,x}\dot{x}$ and $\mu_{l,y}\dot{y}$ vectors.

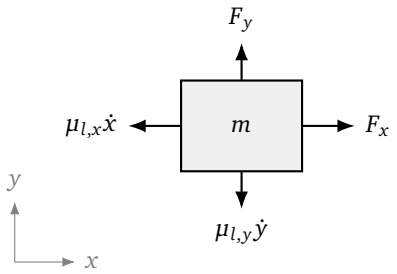


Figure 3: In-plane forces acting on the print head. Friction forces are shown for positive velocities.

Euler-Lagrange The Euler-Lagrange method can be used in order to efficiently derive the mechanical model for the system. The generalized coordinates and velocities are defined as

$$q = \begin{bmatrix} \theta_l \\ \theta_r \end{bmatrix}, \qquad \dot{q} = \begin{bmatrix} \omega_l \\ \omega_r \end{bmatrix}. \tag{11}$$

These states directly determine the x and y velocity of the printer head via Equation (6).

The kinetic energy of the motor and printer head can be combined to determine the K function of Euler-Lagrange

$$K = \underbrace{\frac{1}{2} J_m \left(\omega_l^2 + \omega_r^2\right)}_{\text{Motor}} + \underbrace{\frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2\right)}_{\text{Printer head}}. \tag{12}$$

Substitution of the found kinematics with Equation (6) transforms the equation to

$$K = \frac{1}{2} J_m \left(\omega_l^2 + \omega_r^2\right) + \frac{1}{2} m \left(\left(\frac{r_m}{2}\left(\omega_l + \omega_r\right)\right)^2 + \left(\frac{r_m}{2}\left(\omega_l - \omega_r\right)\right)^2\right), \tag{13}$$

$$L = K = \frac{1}{2} \underbrace{\left(J_m + \frac{m r_m^2}{2}\right)}_{J_{\text{eff}}} \left(\omega_l^2 + \omega_r^2\right). \tag{14}$$

*The derivative is normally denoted by a dot (e.g., $\dot{\theta}$) in this report except when the differential quantity is current (i.e., i). In that case, the explicit notation is used $\left(\text{i.e., } \frac{di}{dt}\right)$ in order to avoid confusion due to the double dot (i.e., $\dot{i}$).

The Lagrangian function is defined by $L = K - P$. As there is no potential energy stored in the model (i.e, the belts are considered non-elastic and constant height) the function $P = 0$. Therefore, $L = K$.

The dissipation function consists of the linear and pulley bearings (i.e., $D = D_{\text{lin}} + D_{\text{pull}}$). The dissipation function for the linear bearings can be modeled and rewritten as

$$D_{\text{lin}} = \frac{1}{2}\mu_{l,x}\dot{x}^2 + \frac{1}{2}\mu_{l,y}\dot{y}^2, \tag{15}$$

$$D_{\text{lin}} = \frac{r_m^2}{8}\left[\mu_{l,x}\left(\omega_l + \omega_r\right)^2 + \mu_{l,y}\left(\omega_l - \omega_r\right)^2\right]. \tag{16}$$

It can be observed from Figure 1 that each motor drives a belt connected to the mass along two fixed pulleys and two pulleys attached to the printer head rail. For the fixed pulley, its rotational speed is equal to $\omega_p = \frac{r_m}{r_p}\omega_l$. Therefore, the dissipation function becomes

$$D_{\text{pul,fixed}} = 2\frac{1}{2}\mu_r\left(r_p\omega_p\right)^2,$$

$$D_{\text{pul,fixed}} = 2\frac{1}{2}\mu_r\left(r_m\omega_l\right)^2.$$

The centers of the four pulleys attached to the printer head rail move in the y-direction. The dissipation term for a pulley with a relative tangential speed v_p is stated as $D_p = \frac{1}{2}\mu_r v_p^2$. The relative tangential speed is equal to $v_p = \dot{x}$. Therefore,

$$D_{\text{pul,rail}} = 4\frac{1}{2}\mu_r\dot{x}^2. \tag{17}$$

Adding the dissipation terms together (i.e, $D = D_{\text{lin}} + D_{\text{pul,fixed,l}} + D_{\text{pul,fixed,r}} + D_{\text{pul,rail}}$) and substitution of Equation (6) leads to

$$D(\omega_l, \omega_r) = \frac{r_m^2}{8}\left[\mu_{l,x}\left(\omega_l + \omega_r\right)^2 + \mu_{l,y}\left(\omega_l - \omega_r\right)^2\right] + \mu_r r_m^2\left(\omega_l^2 + \omega_r^2\right) + \frac{1}{2}\mu_r r_m^2\left(\omega_l + \omega_r\right)^2. \tag{18}$$

All elements of the *Lagrange's equation of motion* are present. From this equation, the complete mechanical model can be derived

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = -\frac{\partial D}{\partial \dot{q}} + \underbrace{\begin{bmatrix} K_m i_{m,l} \\ K_m i_{m,r} \end{bmatrix}}_{\tau_m}. \tag{19}$$

From Equation (14) it follows that

$$\frac{\partial L}{\partial q} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = J_{\text{eff}}\begin{bmatrix} \dot{\omega}_l \\ \dot{\omega}_r \end{bmatrix}. \tag{20}$$

From Equation (18) the dissipation element is determined by

$$\frac{\partial D}{\partial \omega_l} = \underbrace{\frac{r_m^2}{4}\left[\mu_{l,x} + \mu_{l,y} + 12\mu_r\right]}_{d_s}\omega_l + \underbrace{\frac{r_m^2}{4}\left[\mu_{l,x} - \mu_{l,y} + 4\mu_r\right]}_{d_c}\omega_r \tag{21}$$

$$\frac{\partial D}{\partial \omega_r} = \underbrace{\frac{r_m^2}{4}\left[\mu_{l,x} - \mu_{l,y} + 4\mu_r\right]}_{d_c}\omega_l + \underbrace{\frac{r_m^2}{4}\left[\mu_{l,x} + \mu_{l,y} + 12\mu_r\right]}_{d_s}\omega_r. \tag{22}$$

Which can be written compactly as

$$\frac{\partial D}{\partial \dot{q}} = \begin{bmatrix} d_s\omega_l + d_c\omega_r \\ d_c\omega_l + d_s\omega_r \end{bmatrix}. \tag{23}$$

Substitution into Equation (19) results in

$$J_{\text{eff}}\dot{\omega}_l = K_m i_{m,l} - d_s\omega_l - d_c\omega_r, \tag{24}$$
$$J_{\text{eff}}\dot{\omega}_r = K_m i_{m,r} - d_c\omega_l - d_s\omega_r. \tag{25}$$

2.3 State-space model

The states are chosen as

$$\mathbf{x} = \begin{bmatrix} \theta_l, & \theta_r, & \omega_l, & \omega_r, & i_{m,l}, & i_{m,r} \end{bmatrix}^\mathsf{T}. \tag{26}$$

The input vector is defined by

$$\mathbf{u} = \begin{bmatrix} u_{m,l}, & u_{m,r} \end{bmatrix}^\mathsf{T}. \tag{27}$$

The position of the printer head is selected as the output vector of the system

$$\mathbf{y} = \begin{bmatrix} x, & y \end{bmatrix}^\mathsf{T}. \tag{28}$$

This yields the following state-space representation

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{d_s}{J_{\text{eff}}} & -\frac{d_c}{J_{\text{eff}}} & \frac{K_{m,l}}{J_{\text{eff}}} & 0 \\ 0 & 0 & -\frac{d_c}{J_{\text{eff}}} & -\frac{d_s}{J_{\text{eff}}} & 0 & \frac{K_{m,r}}{J_{\text{eff}}} \\ 0 & 0 & -\frac{K_{m,l}}{L_{m,l}} & 0 & -\frac{R_{m,l}}{L_{m,l}} & 0 \\ 0 & 0 & 0 & -\frac{K_{m,r}}{L_{m,r}} & 0 & -\frac{R_{m,r}}{L_{m,r}} \end{bmatrix}}_{\mathbf{A}} \mathbf{x} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_{m,l}} & 0 \\ 0 & \frac{1}{L_{m,r}} \end{bmatrix}}_{\mathbf{B}} \mathbf{u} \tag{29}$$

$$\mathbf{y} = \underbrace{\begin{bmatrix} \frac{r_m}{2} & \frac{r_m}{2} & 0 & 0 & 0 & 0 \\ \frac{r_m}{2} & -\frac{r_m}{2} & 0 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{C}} \mathbf{x}.$$

This model assumes that the

- belts are non-elastic and do not slip,
- motor parameters are equal for the left and right motor,
- mass of the printer head rail, pulleys, and inertia of the pulleys can be neglected,
- parameters $\mu_{l,x}, \mu_{l,y}$ are total linear damping coefficients per direction,
- parameter μ_r denotes the tangential damping per pulley.

Question 3 Decouple the x and y dynamics

3.1 Decoupling equations

One can rewrite Equation (6) by choosing new velocity variables $\dot{x} = v_x, \dot{y} = v_y$. This results in

$$v_x = \frac{r_m}{2}\left(\omega_l + \omega_r\right), \qquad v_y = \frac{r_m}{2}\left(\omega_l - \omega_r\right). \tag{30}$$

When Equations (24) and (25) are added to each other, each side is multiplied with $\frac{r_m}{2}$ and Equation (30) are substituted this leads to

$$J_{\text{eff}}\dot{v}_x = \frac{r_m K_m}{2}\underbrace{\left(i_{m,l} + i_{m,r}\right)}_{i_x} - \left(d_s + d_c\right)v_x. \tag{31}$$

Similarly, for the movement along the y-direction, one can write

$$J_{\text{eff}}\dot{v}_y = \frac{r_m K_m}{2}\underbrace{\left(i_{m,l} - i_{m,r}\right)}_{i_y} - \left(d_s - d_c\right)v_y. \tag{32}$$

The electrical equations can be transformed similarly by adding Equation (10) for the left and subtraction for the right motor

$$L_{m,l}\frac{di_x}{dt} = -R_m i_x - \frac{2K_m}{r_m}v_x + \underbrace{\left(u_{m,l} + u_{m,r}\right)}_{u_x}, \tag{33}$$

$$L_{m,r}\frac{di_y}{dt} = -R_m i_y - \frac{2K_m}{r_m}v_y + \underbrace{\left(u_{m,l} - u_{m,r}\right)}_{u_y}. \tag{34}$$

3.2 Transformed state-space model

In order to decouple the x and y coordinates two state-space models are presented. The two state vectors are chosen as

$$\mathbf{x}_x = \left[x,\ v_x,\ i_x\right]^\mathsf{T}, \qquad \mathbf{x}_y = \left[y,\ v_y,\ i_y\right]^\mathsf{T}. \tag{35}$$

The inputs of the two models are represented by u_x, u_y. The position of the printer head is selected as x, y. Resulting in the transformed state-space representation

x-direction

$$\dot{\mathbf{x}}_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{d_s+d_c}{J_{\text{eff}}} & \frac{r_m K_m}{2J_{\text{eff}}} \\ 0 & -\frac{2K_m}{r_m L_m} & -\frac{R_m}{L_m} \end{bmatrix}\mathbf{x}_x + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix}u_x,$$

$$x = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{x}_x. \tag{36}$$

y-direction

$$\dot{\mathbf{x}}_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{d_s-d_c}{J_{\text{eff}}} & \frac{r_m K_m}{2J_{\text{eff}}} \\ 0 & -\frac{2K_m}{r_m L_m} & -\frac{R_m}{L_m} \end{bmatrix}\mathbf{x}_y + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix}u_y,$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{x}_y. \tag{37}$$

Question 4 Drawing shapes

The corresponding motor voltages need to be determined in order to draw an ellipse with the 3D-printer. The x, and $y-$coordinates can be stated as

$$x_{\text{ref}} = x_c + a\cos(\Omega t), \qquad y_{\text{ref}} = y_c + b\sin(\Omega t). \tag{38}$$

Whereas $a, b \neq 0\,\text{m}$, $a \neq b$ and $\Omega = \frac{2\pi}{T}$. A circle is drawn when $a = b$.

Using the decoupled state-space matrices (i.e., Equations (36) and (37)), the motor voltages can be derived

$$\begin{aligned} J_{\text{eff}}\ddot{x}_{\text{ref}} &= \frac{r_m K_m}{2} i_x - (d_s + d_c)\,\dot{x}_{\text{ref}}, \\ i_{x,\text{ref}} &= \frac{2}{r_m K_m}\left[J_{\text{eff}}\ddot{x}_{\text{ref}} + (d_s + d_c)\,\dot{x}_{\text{ref}}\right], \\ u_x &= L_m \frac{di_{x,\text{ref}}}{dt} + R_m i_{x,\text{ref}} + \frac{2K_m}{r_m}\dot{x}_{\text{ref}}. \end{aligned} \qquad \begin{aligned} J_{\text{eff}}\ddot{y}_{\text{ref}} &= \frac{r_m K_m}{2} i_y - (d_s - d_c)\,\dot{y}_{\text{ref}}, \\ i_{y,\text{ref}} &= \frac{2}{r_m K_m}\left[J_{\text{eff}}\ddot{y}_{\text{ref}} + (d_s - d_c)\,\dot{y}_{\text{ref}}\right], \\ u_y &= L_m \frac{di_{y,\text{ref}}}{dt} + R_m i_{y,\text{ref}} + \frac{2K_m}{r_m}\dot{y}_{\text{ref}}. \end{aligned} \tag{39}$$

When $a = 0.1\,\text{m}$, $b = 0.07\,\text{m}$, $x_c = \frac{dx}{2} = 0.145\,\text{m}$, $y_c = \frac{dy}{2} = 0.145\,\text{m}$ and $T = 10\,\text{s}$ the model returns the following plot given in Figure 4.

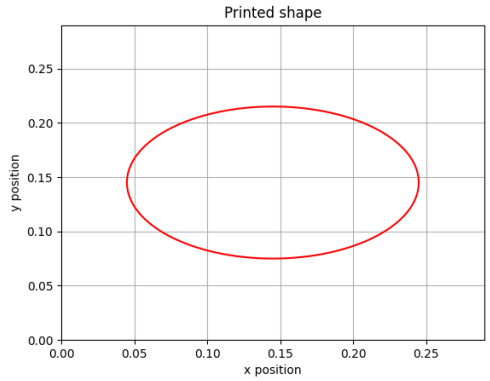


Figure 4: The ellipse drawn for $a = 0.1$ and $b = 0.07$.

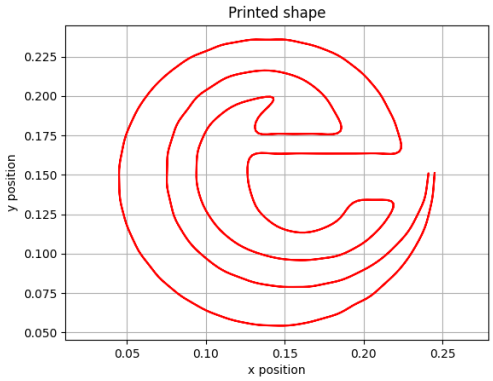


Figure 5: The Efteling logo. Drawn with $k = 250$ in Equation (40).

From Equation (38) it can be observed that using geometric functions an arbitrary shape can be drawn. Using the complex Fourier series one can convert the contour of an image in a mathematical relation. The complex Fourier series and its relation to the printer head position is given by

$$z(t) = \sum_k c_k e^{ik\Omega t}, \qquad x_{\text{ref}}(t) = \text{Re}\{z(t)\}, \qquad y_{\text{ref}}(t) = \text{Im}\{z(t)\}. \tag{40}$$

The website [1] is used to obtain the coefficients c_k. When the coefficients from [2] are used, the 3D-printer draws Figure 5.

Question 5 Model the Nozzle

The lumped model of the nozzle is shown in Figure 6.

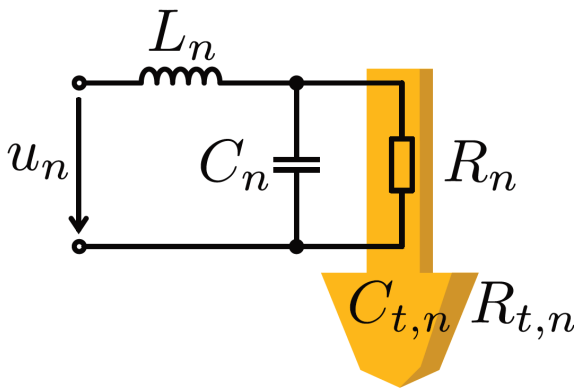


Figure 6: The lumped model of the nozzle in the electrical and thermal domain.

Assuming that all energy dissipated in R_n is converted to heat and defining the voltage across R_n as v_n, one can write

$$P_{\text{heat}} = v_n i_R = \frac{v_n^2}{R_n}. \tag{41}$$

The thermal capacity determines the amount of energy needed to increase the temperature by

$$C_{t,n}\dot{T}_n = P_{\text{heat}} - \frac{T_n - T_e}{R_{t,n}}. \tag{42}$$

Substitution of Equation (41) and solving for $\dot{T}_n$ leads to

$$\dot{T}_n = \frac{1}{C_{t,n}}\left[\frac{v_n^2}{R_n} - \frac{T_n - T_e}{R_{t,n}}\right]. \tag{43}$$

Applying Kirchhoff's voltage and current laws leads to

$$L_n \frac{di_L}{dt} = u_n - v_n, \tag{44}$$

$$C_n \dot{v}_n = i_L - \frac{v_n}{R_n}. \tag{45}$$

A suitable choice for the state vector is

$$\mathbf{x}_n = [T_n,\ i_L,\ v_n]^\mathsf{T}. \tag{46}$$

Transforming into the requested format (i.e, $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$) yields

$$\frac{d}{dt}\begin{bmatrix} T_n \\ i_L \\ v_n \end{bmatrix} = \begin{bmatrix} \frac{1}{C_{t,n}}\left(\frac{v_n^2}{R_n} - \frac{T_n - T_e}{R_{t,n}}\right) \\ \frac{u_n - v_n}{L_n} \\ \frac{i_L - \frac{v_n}{R_n}}{C_n} \end{bmatrix}. \tag{47}$$

Question 6 Printing a circle of filament (PDE)

To aid in the derivation, Figure 7 is created to define the conventions.

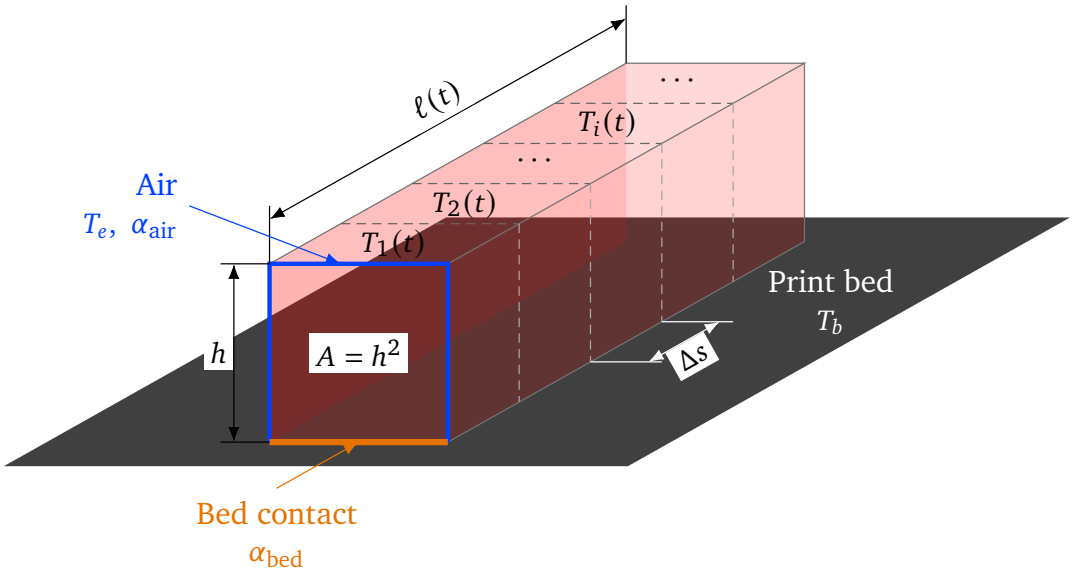


Figure 7: Spatial discretization of the deposited filament. Blue boundaries exchange heat with the air, while the orange boundary exchanges heat with the print bed. The axial heat current is neglected before the printing trajectory is closed.

For a distance along the printed path, it can be defined

$$0 \leq s \leq \ell(t), \qquad \dot{\ell}(t) = v_{\text{print}}(t). \tag{48}$$

The printing speed is given by

$$v_{\text{print}} = \sqrt{\dot{x}^2 + \dot{y}^2}. \tag{49}$$

The functions used for x and y are given in Equation (38).

The cross-sectional area between two pieces of filament is given by $A = h^2$. The boundary between the air and the filament are given by $p_{\text{air}} = 3h$. The print bed and the filament are in contact $p_{\text{bed}} = h$.

The Partial Differential Equation (PDE) for the heat transfer can be modelled as

$$\underbrace{\rho C_p A \frac{\partial T(s,t)}{\partial t}}_{\text{Heat storage}} = \underbrace{\lambda A \frac{\partial^2 T(s,t)}{\partial s^2}}_{\text{Conduction}} - \underbrace{\alpha_{\text{air}} 3h \left(T(s,t) - T_e\right)}_{\text{Convection}} + \underbrace{\alpha_{\text{bed}} h \left(T_b - T(s,t)\right)}_{\text{Heated bed}}. \tag{50}$$

The initial value condition can be found by defining $t_d(s)$ which indicates the time when the filament at position s is printed. At that time and position, the temperature of the filament is equal to the nozzle temperature $T(s, t_d(s)) = T_n$.

The printing temperature of the nozzle is $T_n = 240\,°\text{C}$. A suitable choice for filament is ABS as this temperature falls in the generic print temperature [3]. The material parameters of [3] are used in the PDE evaluation. The effective heat contact coefficient α_{bed} is selected as $50\,\text{W}/(\text{m}^2\,\text{K})$ which is found to be adequate to illustrate the progression of the PDE versus time. The filament temperature is considered uniform when $\max_i T_i - \min_i T_i \leq 1\,\text{K}$.

Table 1. Material input parameters [26].

Material	Density ρ (kg/m^3)	Specific Heat C_P (J/kg·K)	Thermal Conductivity λ (W/m·K)	Length L (mm)	Thickness ϕ (mm)	Height h (mm)
ABS	1050	2080	0.177	50	~0.7	~0.2

h_{ABS} (ABS convective heat transfer coefficient considered in the simulations) = 250 (W/m^2·K).

Figure 8: The ABS material properties as presented in [3]. These values are treated as generic parameters for ABS filament and discrepancies between different colors may be present.

In order to evaluate Equation (50) one can use the Method of Lines (MOL) in order to numerically compute the solution by Python. The derivative is approximated by means of the *Forward Euler Method* (i.e, $\frac{du_i}{dt} \approx \frac{u_i^{n+1}-u_i^n}{\Delta t}$). Substitution in Equation (50) leads to

$$\frac{dT_i}{dt} \approx \underbrace{\frac{\lambda}{\rho C_p}}_{a_{\text{th}}} \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta s^2} - \underbrace{\frac{3\alpha_{\text{air}}}{\rho C_p h}}_{\beta_{\text{air}}} (T_i - T_e) + \underbrace{\frac{\alpha_{\text{bed}}}{\rho C_p h}}_{\beta_{\text{bed}}} (T_b - T_i)\,. \tag{51}$$

The PDE is numerically evaluated using Python. The x and y-coordinates as a function of time and temperature of the filament are presented in the animation of Figure 9.

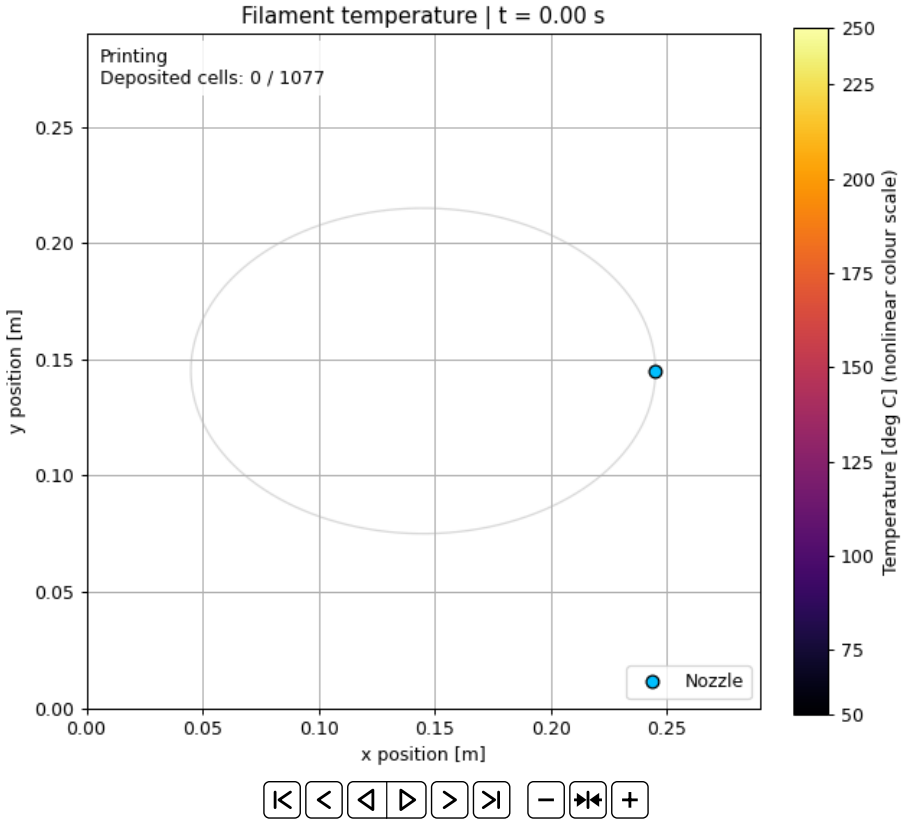


Figure 9: Printing and cooling of the filament. Open in Acrobat Reader in order to start the animation.

As time progresses, the temperature of the filament reaches a stationary value of

$$T_\infty = \frac{3\alpha_{\text{air}} T_e + \alpha_{\text{bed}} T_b}{3\alpha_{\text{air}} + \alpha_{\text{bed}}} = \frac{3 \cdot 25 \cdot 300 + 50 \cdot 378.15}{3 \cdot 25 + 50} = 331.26\,\text{K} = 58.11\,^\circ\text{C}. \tag{52}$$

In Figure 10 the temperature versus time graph of the first and last segment are presented. When the ellipse closes, the first and last segment come into contact. The last element initially heats up the first segment after which the temperature of both segments cools down to the stationary value calculated in Equation (52).

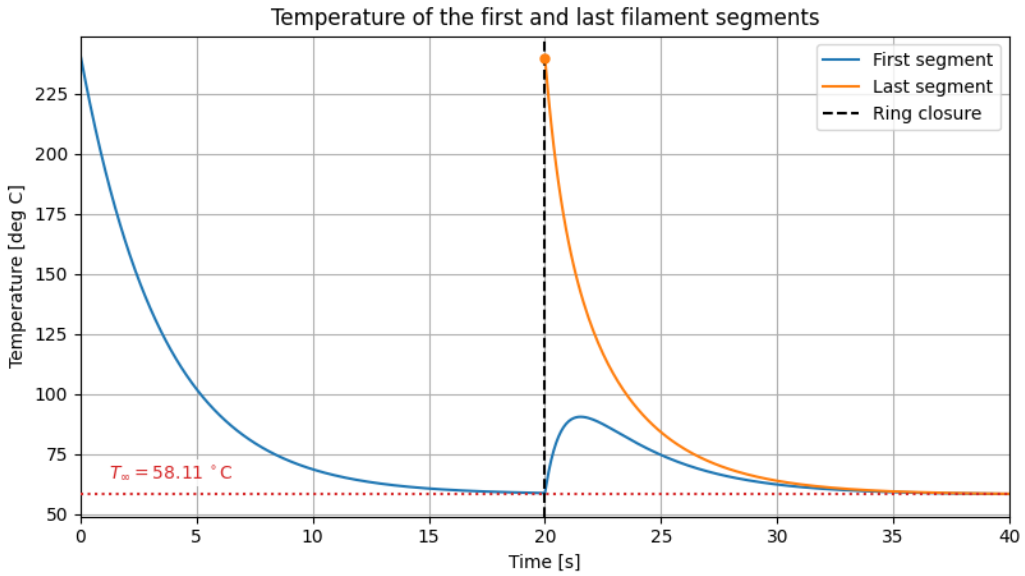


Figure 10: Temperature of the first and last segment of the printed ellipse. When the ring is closed, the first element heats up due to the thermal conductivity of the filament.

2 Finite State Machine

Question 1 Finite State Machines — Filament replacement

The Finite State Machine (FSM) is presented in Figure 11. The definitions of the states are given in Table 2. The meaning of the events are summarized in Table 3.

Table 2: Controller states.

State	Description
q_0	Waiting for a new print to be loaded
q_1	Heating the nozzle
q_2	Return to (saved) printing location
q_3	Printing the design
q_4	Moving to the wiping location $(x, y) \to (0, 0)$.
q_5	Wiping the excess filament off the nozzle
q_6	Loading new (colored) filament
q_7	Move to parking location

Table 3: Controller events.

Event	Guard
$e_{\text{new,print}}$	New printing job
e_{hot}	Nozzle reaches printing temperature $T_n \geq T_{\text{ref}} - 1\,\text{K}$
e_{return}	Reaches (saved) printing position
e_{replace}	Color change or finishes filament $(C \vee F)$
e_{park}	Reached parking position $(x, y) \to (0, 0)$.
e_{wipe}	Nozzle is cleaned
e_{load}	New filament is loaded
$e_{\text{print,complete}}$	Print is finished

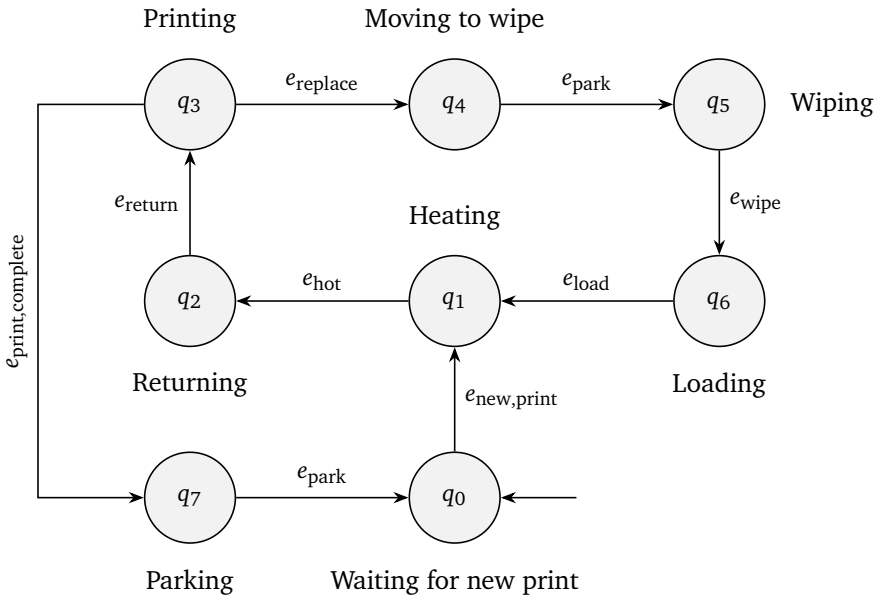


Figure 11: Extended finite state machine of the 3D-printer. The initial state is q_0.

The systems start in q_0 and is denoted as the initial condition. The system waits for a new printing design to be loaded. When a new printing design is received, the system transitions to the heating state q_1. In this stage, the nozzle is heated to a given reference temperature T_{ref}. When this temperature is reached within 1 K, the printer head can be moved towards the printing position. In the case of a saved position, the printer head will move to the saved state. This decision is taken in q_2. When the printer head has reached the desired position, the system enters the printing q_3 stage. The printer will continue printing until

1. A change of color is requested, or the filament roll is finished ($C \vee F$).
2. The printing trajectory is completed.

In the case of a control flow signal being present (i.e., $C \vee F = 1$) the system will save its current position and move to the wiping position ($(x, y) \rightarrow (0, 0)$) denoted by stage q_4. At that position, the nozzle will be wiped in order to remove any excess filament coming out of the nozzle in state q_5. In this stage, the nozzle is turned off prior to the loading of new filament. After loading, the nozzle is turned on again to resume printing and reaches state q_1 again.

When the printed design has completed. The system will move towards the parking state q_7 which will move the printing head to the parking position ($(x, y) \rightarrow (0, 0)$). When the printer head reaches the parking position, the system reaches q_0 which is the initial stage of the printer.

Question 2 Automata — Parked detection

The automaton for the parked detection is given in Figure 12.

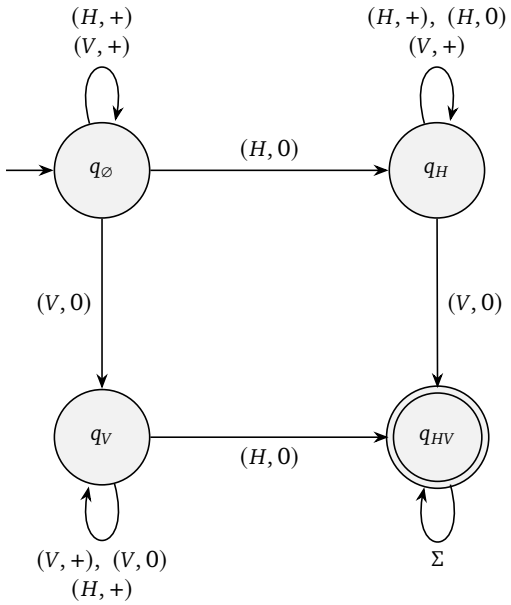


Figure 12: Automaton for detecting the bottom-left parking position. The accepting state q_{HV} indicates that both end positions have been reached.

Table 4: Automaton states.

State	Description
$q_\varnothing$	Printer head has not reached a boundary
q_H	Printer head has reached the left boundary
q_V	Printer head has reached the bottom boundary
q_{HV}	Printer head has reached the origin.

The parking detector is defined by

$$\mathcal{A} = (Q, \Sigma, \delta, q_\varnothing, F_{\text{acc}}), \tag{53}$$

where

$$Q = \{q_\varnothing, q_H, q_V, q_{HV}\}, \qquad \Sigma = \{(H,+), (H,0), (V,+), (V,0)\}, \qquad F_{\text{acc}} = \{q_{HV}\}. \tag{54}$$

Element H denotes the horizontal movement of the printer head whereas V denotes the vertical movement. A + indicates that the printer heads velocity is non-zero and is therefore moving. When a 0 is reported, the printer head has stopped moving and has reached the boundary of the printer bed. The parked state is reached after both $(H,0) \wedge (V,0)$ have been confirmed during the return to home procedure.

3 Hybrid system

Question 1 Filament replacement

The mechanical, electrical and nozzle models derived in the previous chapters are combined into the following continuous dynamics.

1.1 Model

The physical state vector is defined as

$$\mathbf{z} = \left[x\ , v_x\ , i_x\ , y\ , v_y\ , i_y\ , T_n\ , i_L\ , v_n\right]^\mathsf{T}. \tag{55}$$

The coupled continuous dynamics are given by

$$\dot{\mathbf{z}} = \begin{bmatrix} A_x \mathbf{x}_x + B_x u_x \\ A_y \mathbf{x}_y + B_y u_y \\ f_n(\mathbf{x}_n, u_n) \end{bmatrix}, \qquad \mathbf{u} = \kappa(q, \mathbf{z}, \tau_p) \tag{56}$$

where τ_p denotes the effective printing time and $\mathbf{u} = \begin{bmatrix} u_x, & u_y, & u_n \end{bmatrix}^\mathsf{T}$. The operating mode determines the voltage level that needs to be applied to the motors. During the printing state, the motor voltages are calculated from the feedforward expressions given in Equation (39). In the parking and returning stages the horizontal and vertical movements are performed after each other as given in Figure 12. The nozzle voltage is switched off during loading and waiting. The voltage is switched on during heating, printing, and wiping.

$$\dot{\tau}_p = \begin{cases} 1, & q = \text{PRINTING}, \\ 0, & \text{otherwise}, \end{cases} \qquad \dot{\tau}_q = 1, \tag{57}$$

The timer for the mode duration τ_q is reset to zero when an event has occurred. The effective printing timer τ_p continues only during the printing stage. Its status is preserved during other states.

1.2 Numerical parameters

For the nozzle temperature dynamics given in Equation (47) the parameters given in Table 5 are used.

Table 5: Additional parameters used in the hybrid simulation.

Parameter	Value	Basis
L_n	10 mH	Assumed electrical parameter.
C_n	100 µF	Assumed electrical parameter.
$C_{t,n}$	4.55 J/K	Estimated using $C_{t,n} = m_n c_p$.
$u_{n,\text{eq}}$	13.76 V	Calculated from the thermal equilibrium.
t_{wipe}	50 s	Assumed duration of the wiping process.
T_{ref}	240 °C	Selected printing temperature.

The integration of the nozzle model is done using the `Radau` method [4]. The nozzle is assumed to be made of brass. The material properties in order to model the nozzle are given in [5].

1.3 Guards

Table 6: Guards denoting the conditions that need to be met in order to initiate an event.

Transition	Condition
$q_0 \rightarrow q_1$	New printing job available.
$q_1 \rightarrow q_2$	$T_n \geq T_{\text{ref}} - 1\,\text{K}$.
$q_2 \rightarrow q_3$	Goal position reached after returning.
$q_3 \rightarrow q_4$	$C \vee F$
$q_4 \rightarrow q_5$	Printer head reaches the origin $(x, y) \rightarrow (0, 0)$.
$q_5 \rightarrow q_6$	$\tau_q \geq t_{\text{wipe}}$.
$q_6 \rightarrow q_1$	Considered when $T_n \leq 333\,\text{K}$.
$q_3 \rightarrow q_7$	Print trajectory finished.
$q_7 \rightarrow q_0$	Printer head reaches the origin $(x, y) \rightarrow (0, 0)$.

For this simulation, filament loading is assumed to be completed when the nozzle temperature has decreased to 333 K.

1.4 Reset

In case of a replacement request ($C \vee F = 1$), the mechanical state is saved, and the timer τ_q is reset at a mode transition

$$\mathbf{z}^{+}_{\mathrm{s,mech}} = \left[x\ ,\ v_x\ ,\ i_x\ ,\ y\ ,\ v_y\ ,\ i_y\right]^{\mathsf{T}}. \qquad \tau_q^{+} = 0. \tag{58}$$

When printing resumes, the saved mechanical state is restored

$$\mathbf{z}^{+}_{\mathrm{mech}} = \mathbf{z}_{\mathrm{s,mech}}. \tag{59}$$

An idealized reset is simulated by restoring the saved velocities and current used to reconnect the simulation of the printing trajectory.

1.5 Invariants

Table 7: Mode invariants and admissible operating conditions.

Invariant	Condition
q_0	Printing head remains parked
q_1	$T_n \leq T_{\mathrm{ref}} - 1\,\mathrm{K}$
q_2	No additional invariant
q_3	$\neg(C \vee F)$ and $0 \leq \tau_p \leq 500\,\mathrm{s}$
q_4	No additional invariant
q_5	$0 \leq \tau_q \leq t_{\mathrm{wipe}}$
q_6	$T_n \geq 333\,\mathrm{K}$.
q_7	No additional invariant

1.6 Simulation settings

1. The PDE of the thermal dynamics of the filament is updated using Forward Euler.
2. `Radau` integrates the coupled electromechanical and nozzle dynamics.
3. New filament is deposited only during printing and is initialized at the current nozzle temperature.
4. A color change request interrupts the printing process.
5. After resuming, the remaining trajectory is finished.
6. After parking, the printer waits with input $u_n = 0$.

The calculated 58.11 °C is connected to the filament while in contact with the heated bed. The nozzle is not in direct contact with the heated bed and therefore settles to the room temperature of 26.85 °C.

Question 2 Draw a square

The animation Figure 13 demonstrates the total functioning of the 3D-printer. The simulation starts with the nozzle temperature initially equal to the room temperature and the printers head positioned in the parking state. At first the nozzle heats up. Printing of the square starts when the nozzle is heated up. The temperature of the filament is shown, and it can be observed that the newly printed filament is heated to the nozzle temperature. The filament cools down to the equilibrium temperature. After $\tau_p = \frac{500}{3} \approx 166.7\,\mathrm{s}$ of effective printing time, a new color is requested and the printer head will return to the origin and stay there for 50 s to simulate the wiping action. After the wiping action, the nozzle cools down to 333 K simulating that new filament is loaded. When the filament is loaded, the nozzle heats up again to the reference temperature and the printer head moves to the saved position. When this position is reached, the printer continues to complete the print trajectory. After finishing, the printer head moves to the parked position and the nozzle is turned off.

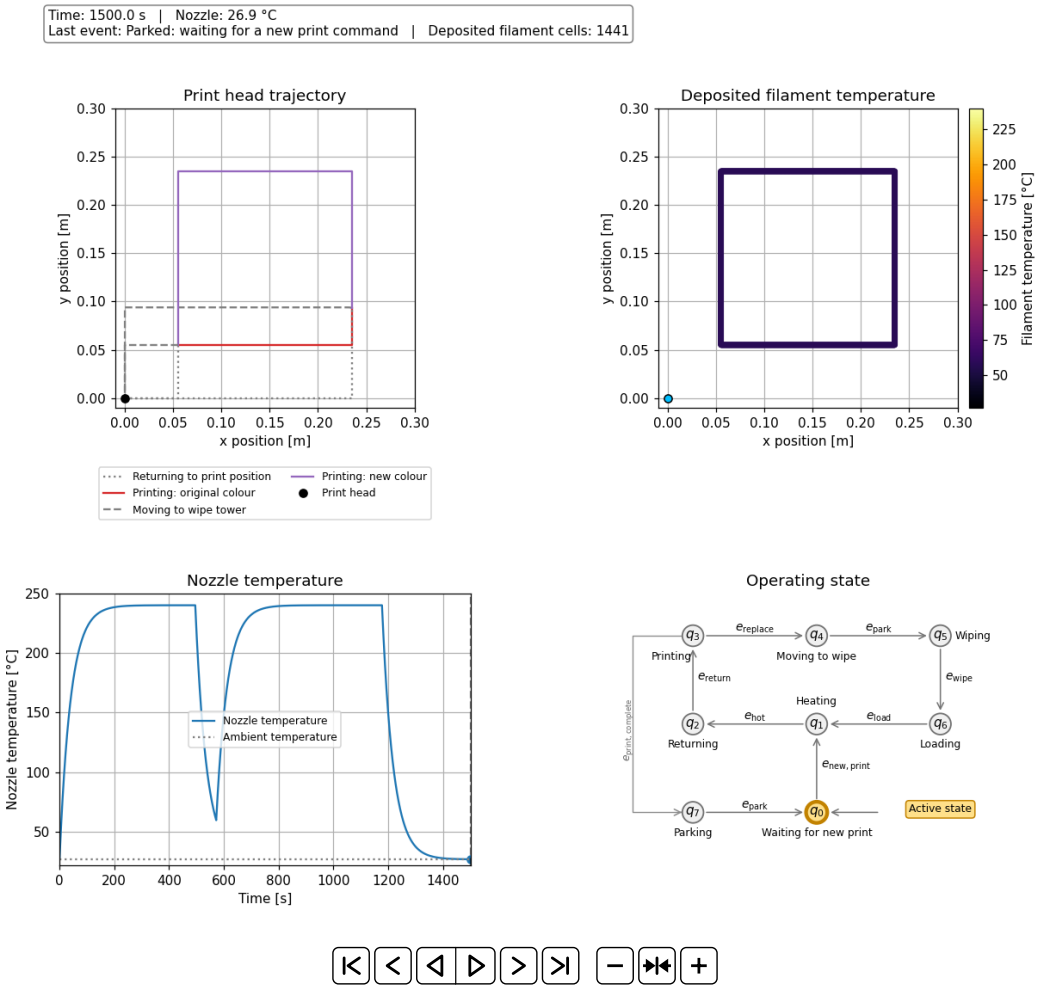


Figure 13: Simulation of the hybrid 3D-printer model. Open in Acrobat Reader in order to start the animation.

The angle speed sensors can be used to determine the position of the printer head with the equations

$$\dot{\hat{x}} = \frac{r_m}{2}\left(\omega_l + \omega_r\right), \qquad \dot{\hat{y}} = \frac{r_m}{2}\left(\omega_l - \omega_r\right). \tag{60}$$

By integration, the position of the printer head $\hat{x}, \hat{y}$ can be obtained. A square has four straight sides by which the velocity of $\dot{\hat{x}}$ and $\dot{\hat{y}}$ will be zero sequentially. The movement of the printer head slows down when it reaches a corner of the square and is equal to zero at the corner. When the velocity of the printer head is equal to zero, the next side of the square will be drawn. The position estimates are obtained by integrating the measured motor angular velocities from a known initial position. The next side is started only after the estimated corner position has been reached, and the measured speed is approximately zero. Ideal sensors and no belt slip are assumed.

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